

Cat vs Dog Image Classification using CNN

PROJECT OVERVIEW

Objective

- Developed an end-to-end Computer Vision application to classify images as either **Cat** or **Dog** using a Convolutional Neural Network (CNN).
- Built a complete deep learning pipeline from data preprocessing and model training to deployment through a user-friendly Streamlit web application.

Tech Stack

Python

TensorFlow / Keras

NumPy, Pandas,
Matplotlib

Streamlit

VS Code & Jupyter Notebook

Workflow & Model Architecture

Project Workflow

01

Dataset Loading (Images)

02

Image Preprocessing

- Resize images
- Normalize pixel values
- Data Augmentation

03

CNN Model Development

- Convolution Layers
- Max Pooling Layers
- Flatten Layer
- Dense Layers
- Sigmoid Output Layer

04

Model Training & Validation

05

Performance Evaluation

06

Deployment using Streamlit

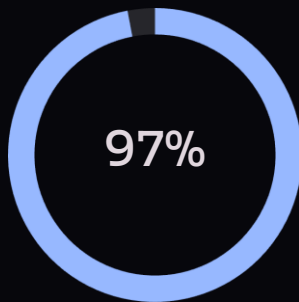
Implementation & Results

Model Performance

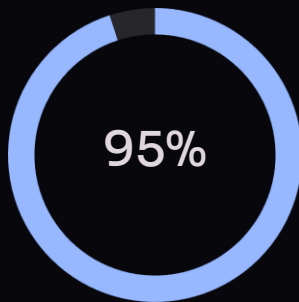
Training Process

- Optimizer: Adam
- Loss Function: Binary Crossentropy
- Accuracy Metric
- Trained on approximately 20,000 images

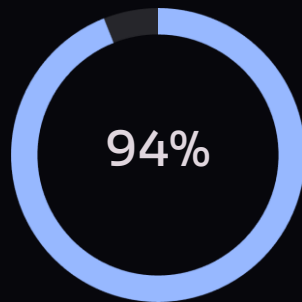
Results



Training Accuracy



Validation Accuracy



Test Accuracy

Sample Prediction

Dog 🐕

Confidence: 97%

Cat 🐈

Confidence: 96%

Deployment & Key Features

Streamlit Web Application

Features

- Upload any Cat or Dog image (.jpg/.png)
- Automatic image preprocessing
- Real-time prediction
- Confidence score display
- Clean and interactive user interface

Benefits

Easy to use

Fast prediction

Demonstrates complete end-to-end deployment of a Deep Learning model

Learning Outcomes & Future Scope

Key Learnings

- Implemented CNN architecture for image classification.
- Applied image preprocessing and data augmentation techniques.
- Improved model performance through hyperparameter tuning.
- Learned end-to-end Deep Learning project development and deployment using Streamlit.

Future Enhancements

- 1 Integrate Grad-CAM for model explainability.
- 2 Deploy on Streamlit Cloud or Hugging Face Spaces.
- 3 Extend to multi-class animal classification.
- 4 Improve prediction reliability using confidence thresholding.